Wave Reflection and Transmission at Normal Incidence – Part 1

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

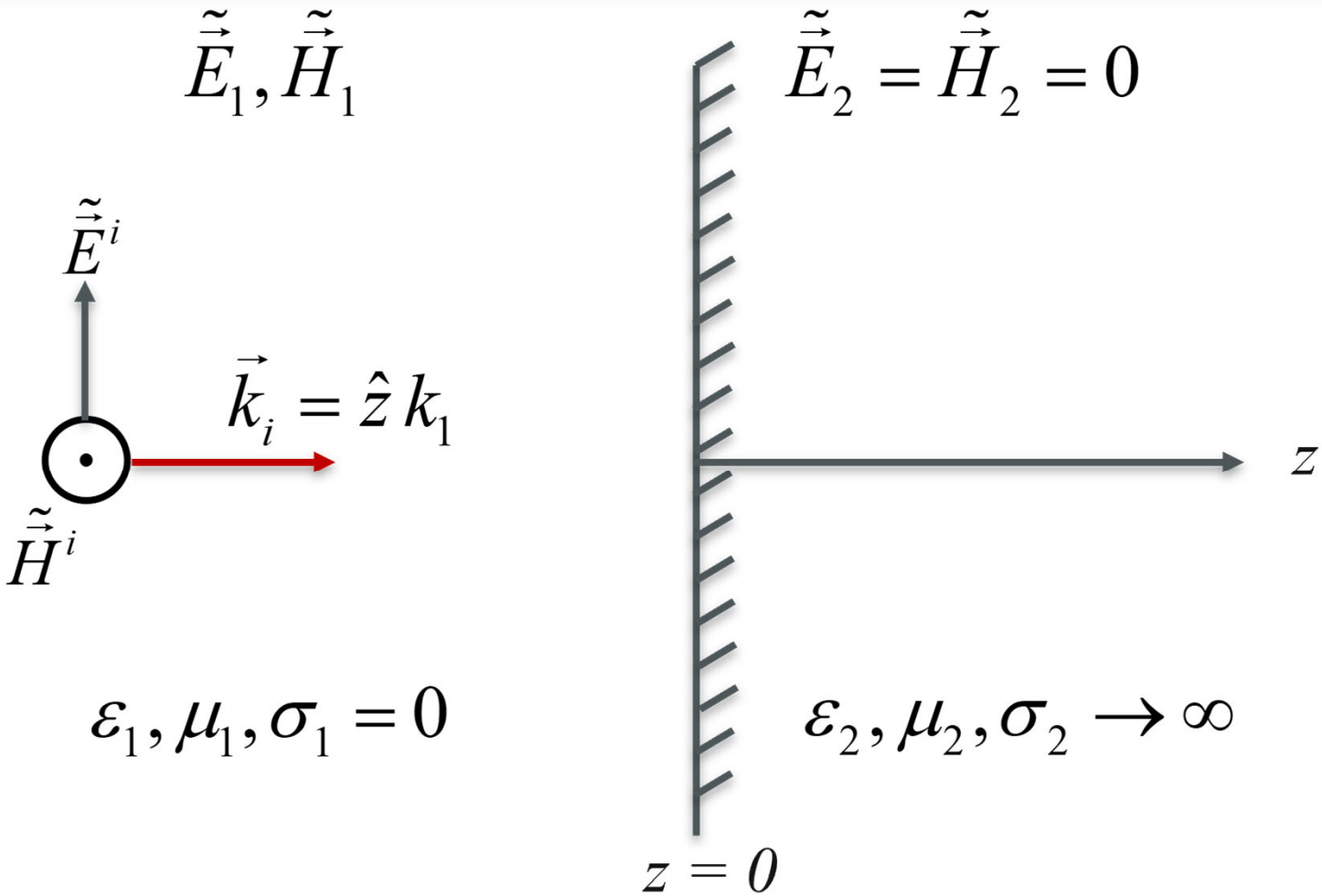
IRA A. FULTON SCHOOLS OF engineering
ARIZONA STATE UNIVERSITY

Lecture Objective

By the end of this lecture, you will be prepared to:

- Analyze plane wave reflection from a perfectly conducting boundary at normal incidence.

Normal Incidence On a Plane Perfectly Conducting Boundary



Incident Plane Wave

- In the absence of the obstacle (medium 2), assume an ***incident*** field is established by the source and is given by

$$\tilde{\vec{E}}^i(z) = \hat{x} E_0^i e^{-jk_1 z}$$

- And the corresponding magnetic field is given by

$$\tilde{\vec{H}}^i(z) = \frac{1}{\eta_1} \hat{z} \times \tilde{\vec{E}}^i = \hat{y} \frac{E_0^i}{\eta_1} e^{-jk_1 z}$$

- Where $k_1 = \omega\sqrt{\mu_1 \varepsilon_1}$ $\qquad \eta_1 = \sqrt{\frac{\mu_1}{\varepsilon_1}}$

Total Field in Region 1

- With the obstacle present, the field in region 1 must satisfy both Maxwell's equations (or equivalently, the Helmholtz equation) and the boundary condition at $z = 0$.
- Since the source excites the incident plane wave in the absence of the obstacle, and the boundary is an infinite plane, it makes sense to assume that the field in region 1 is a superposition of plane waves.

Total Field in Region 1 Continued

- Assume that the total field in region 1 is equal to the incident wave plus a "reflected" wave:

$$\tilde{\vec{E}}_1(z) = \tilde{\vec{E}}^i(z) + \tilde{\vec{E}}^r(z)$$

- Where

$$\tilde{\vec{E}}^r(z) = \hat{x}E_0^r e^{+jk_1 z}$$

- The reflected wave satisfies the Helmholtz equation in region 1.
- Physically, the reflected wave is propagating in the –z-direction (back toward source).

$$\vec{k}_r = -\hat{z}k_1$$

Reflected Plane Wave

- The amplitude of the reflected plane wave follows from the boundary condition:

$$\tilde{\vec{E}}_1(z=0)=0$$

- Thus

$$E_r^0=-E_i^0$$

- And the corresponding magnetic field is given by

$$\tilde{\vec{H}}^r(z)=\frac{1}{\eta_1}\hat{k}_r\times\tilde{\vec{E}}_r=-\frac{1}{\eta_1}\hat{z}\times\tilde{\vec{E}}_r=-\hat{y}\frac{E_0^r}{\eta_1}e^{+jk_1z}=\hat{y}\frac{E_0^i}{\eta_1}e^{+jk_1z}$$

Total Field in Region 1 Continued

- The total electric field in region 1 is given by:

$$\tilde{\vec{E}}_1(z) = \tilde{\vec{E}}^i(z) + \tilde{\vec{E}}^r(z) = \hat{x}E_i^0\left(e^{-jk_1z} - e^{+jk_1z}\right)$$

$$= -\hat{x}j2E_0^i\sin(k_1z)$$

- The total magnetic field in region 1 is given by:

$$\tilde{\vec{H}}_1(z) = \tilde{\vec{H}}^i(z) + \tilde{\vec{H}}^r(z) = \hat{y}\frac{E_0^i}{\eta_1}\left(e^{-jk_1z} + e^{+jk_1z}\right)$$

$$= \hat{y}2\frac{E_0^i}{\eta_1}\cos(k_1z)$$

Total Time-Domain Field in Region 1

- The total time-domain electric field in region 1 is given by:

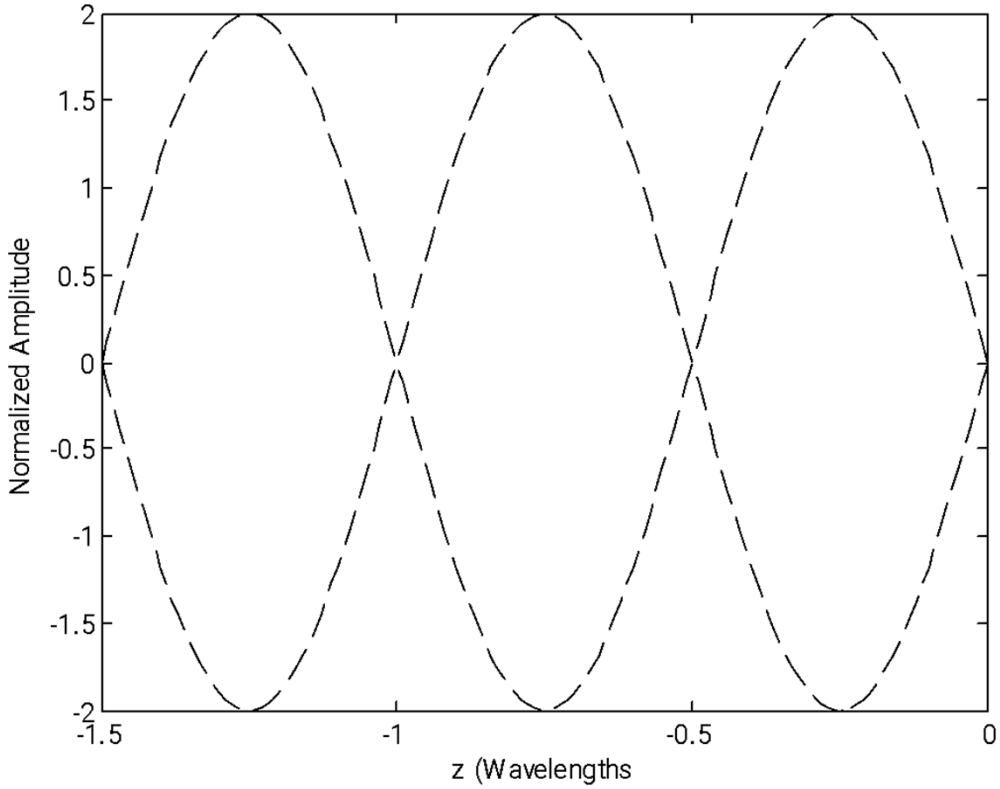
$$\vec{E}_1(z,t) = \mathrm{Re}\left\{-\hat{x}j2E_0^i \sin(k_1 z)\, e^{j\omega t}\right\}$$
$$= \hat{x}2E_0^i \sin(k_1 z)\sin(\omega t)$$

- The total magnetic field in region 1 is given by:

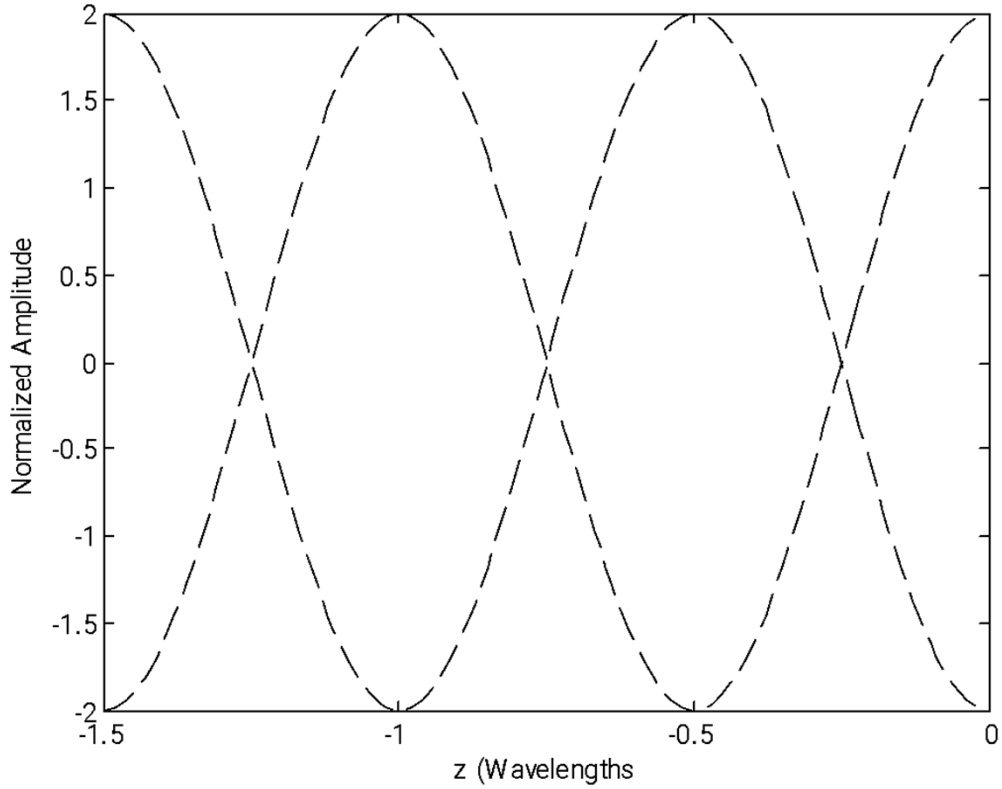
$$\vec{H}_1(z,t) = \mathrm{Re}\left\{\hat{y}2\frac{E_0^i}{\eta_1}\cos(k_1 z)\, e^{j\omega t}\right\}$$
$$= \hat{y}2\frac{E_0^i}{\eta_1}\cos(k_1 z)\cos(\omega t)$$

"Standing waves"

Standing Wave Pattern for the Electric Field



Standing Wave Pattern for the Magnetic Field



Summary

- A plane wave normally incident on the planar boundary between a dielectric half-space and a perfect conductor is reflected back into the dielectric medium.
- The total field in the dielectric medium can be interpreted as a standing wave.

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